# MI-24: the Schatten norm complement follows by convexity

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The full Schatten norm inequality in MI-24 follows from the published Heron norm inequality of Dinh, Dumitru, and Franco, a positive matrix comparison already used by Ghabries, Abbas, Mourad, and Assi, and the triangle inequality. The last step is a general convexity observation: if one endpoint of a segment has norm at most the midpoint's norm, then the other endpoint has norm at least the midpoint's norm.

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## The full statement

Let $A, B \in \mathbb{C}^{n\times n}$ be positive definite, where $n \geq 1$, and define

$$G = A^{1/2}(A^{-1/2}BA^{-1/2})^{1/2}A^{1/2},$$
$$L = A^{1/2}(B^{1/2}A^{-1}B^{1/2})^{1/2}A^{1/2}.$$

All square roots are the positive definite square roots. For $1 \leq p < \infty$, write $\|M\|_p = (\operatorname{tr}|M|^p)^{1/p}$, where $|M| = (M^*M)^{1/2}$; $\|M\|_\infty$ is the largest singular value.

### Theorem 1

For every such pair and every $1 \leq p \leq \infty$,

$$\boxed{\|A + B + G + L\|_p \leq \|A + B + 2L\|_p.}$$

This resolves the complete target of MI-24, equivalently Ghabries–Abbas–Mourad–Assi, Conjecture 4.1, including the previously unproved exponents. Neither commutativity nor symmetry of $L(A, B)$ in its arguments is assumed.

## The two comparisons

Put $H = A + B$ and $K = A^{1/2}B^{1/2} + B^{1/2}A^{1/2}$. The published Heron norm inequality of Dinh–Dumitru–Franco, Theorem 4, p. 2 of the author manuscript, gives

$$\|H + 2G\|_p \leq \|H + K\|_p \qquad (1 \leq p < \infty). \tag{1}$$

The same inequality at $p = \infty$ follows by letting $p \to \infty$ in finite dimension. This is the only non-elementary external theorem needed below; it is also recalled in equation (22) and its following paragraph in Ghabries–Abbas–Mourad–Assi.

The second comparison is

$$K \preceq G + L. \tag{2}$$

For completeness, it has the following direct proof. Set $R = A^{-1/2}B^{1/2}$ and take its polar decomposition $R = U|R|$, where $U$ is unitary. Since $|R^*| = U|R|U^*$,

$$|R^*| + |R| - R - R^* = (U - I)|R|(U^* - I) \succeq 0.$$

Congruence by $A^{1/2}$ gives (2), because

$$A^{1/2}|R^*|A^{1/2} = G, \quad A^{1/2}|R|A^{1/2} = L,$$
$$A^{1/2}(R + R^*)A^{1/2} = K.$$

Moreover, $H + K = (A^{1/2} + B^{1/2})^2 \succeq 0$. Hence (2) implies

$$0 \preceq H + K \preceq H + G + L.$$

The min–max principle orders every eigenvalue of these positive semidefinite matrices, so

$$\|H + K\|_p \leq \|H + G + L\|_p \qquad (1 \leq p \leq \infty). \tag{3}$$

This also reproduces the particular comparison in Ghabries–Abbas–Mourad–Assi, Corollary 4.1, without requiring their block-matrix proof as an additional premise.

## The convexity step

**Proof of Theorem 1.** Fix any $p \in [1, \infty]$ and set

$$X = H + 2G, \qquad Y = H + G + L, \qquad Z = H + 2L.$$

Equations (1) and (3) give $\|X\|_p \leq \|Y\|_p$. Also $X + Z = 2Y$. The triangle inequality therefore gives

$$2\|Y\|_p = \|X + Z\|_p \leq \|X\|_p + \|Z\|_p \leq \|Y\|_p + \|Z\|_p.$$

Subtracting $\|Y\|_p$ proves $\|Y\|_p \leq \|Z\|_p$, as claimed. □

The proof covers all parameters and dimensions in the canonical statement. Its contribution is the convexity deduction connecting the two existing comparisons; no new proof of the published Heron inequality or historical priority certification is claimed.

## References


1. **Dinh–Dumitru–Franco (2017).** T. H. Dinh, R. Dumitru, and J. A. Franco. *On a conjecture of Bhatia, Lim and Yamazaki*. Linear Algebra and its Applications **532** (2017), 140–145. Theorem 4, equation (4), p. 2 of the author manuscript. DOI: 10.1016/j.laa.2017.06.040.

2. **Ghabries–Abbas–Mourad–Assi (2021/2022).** M. M. Ghabries, H. Abbas, B. Mourad, and A. Assi. *New log-majorization results concerning eigenvalues and singular values and a complement of a norm inequality*. Section 4: equation (22), Corollary 4.1, Theorem 4.2 and Conjecture 4.1, pp. 11–13 of arXiv:2105.13356v1. Journal DOI: 10.1080/03081087.2022.2059050.

3. **MI-24.** ajt60gaibb/OpenProblemsInNLA. *Schatten norm complement involving Lin's positive definite quantity*. Canonical problem statement. The original mathematical target and permanent identifier are retained.